

Home Lab Project Week 4, Day 3

Task:

Install and configure Wireshark on the Kali Linux VM and prepare the environment for capturing network traffic from the cybersecurity lab.

For **Week 4, Day 3**, the focus is setting up **Wireshark in Kali Linux** so you're ready to capture and analyze traffic in the lab. This builds directly on Day 1's network verification and Day 2's Linux command-line preparation.

Week 4 — Day 3

Task: Install and configure Wireshark on the Kali Linux VM and prepare the environment for capturing network traffic from the cybersecurity lab.

Note: Wireshark comes preinstalled in Kali linux virtual machine.

What you should do

1. Update Kali's package information

```
sudo apt update
```

2. Install Wireshark

If wireshark needs to be installed, the command to type in Kali terminal is:

```
sudo apt install wireshark
```

If Kali asks whether non-root users should be allowed to capture packets, select **Yes**.

3. Add your Kali user to the Wireshark group

```
sudo usermod -a -G wireshark $USER
```

Then log out and back in, or restart Kali, so the group membership takes effect.

You can verify it with:

```
groups
```

Look for:

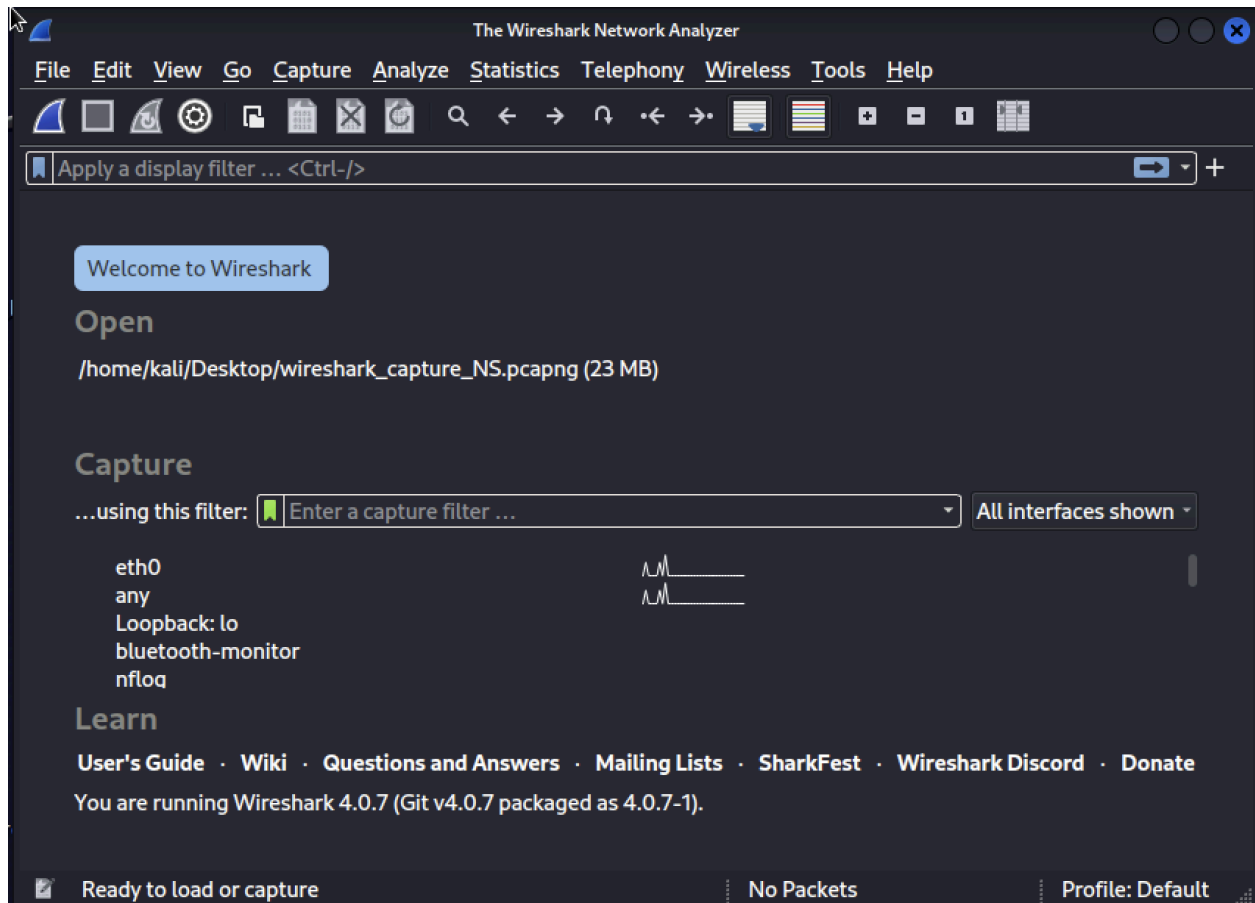
```
wireshark
```

4. Launch Wireshark

From the Kali terminal:

```
wireshark
```

Or launch it from the Kali applications menu.



5. Identify the correct network interface

In Wireshark, look at the available interfaces.

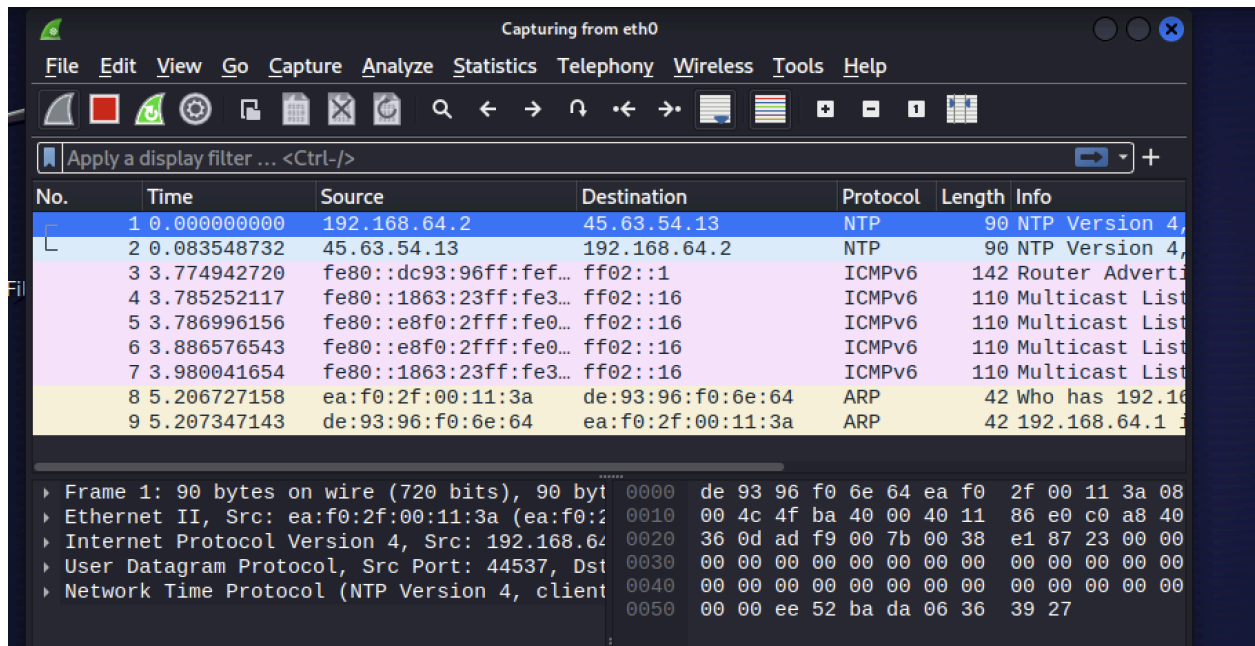
Because your lab uses the isolated **192.168.64.0/24** network, identify the interface associated with Kali's:

```
192.168.64.2
```

You can also use:

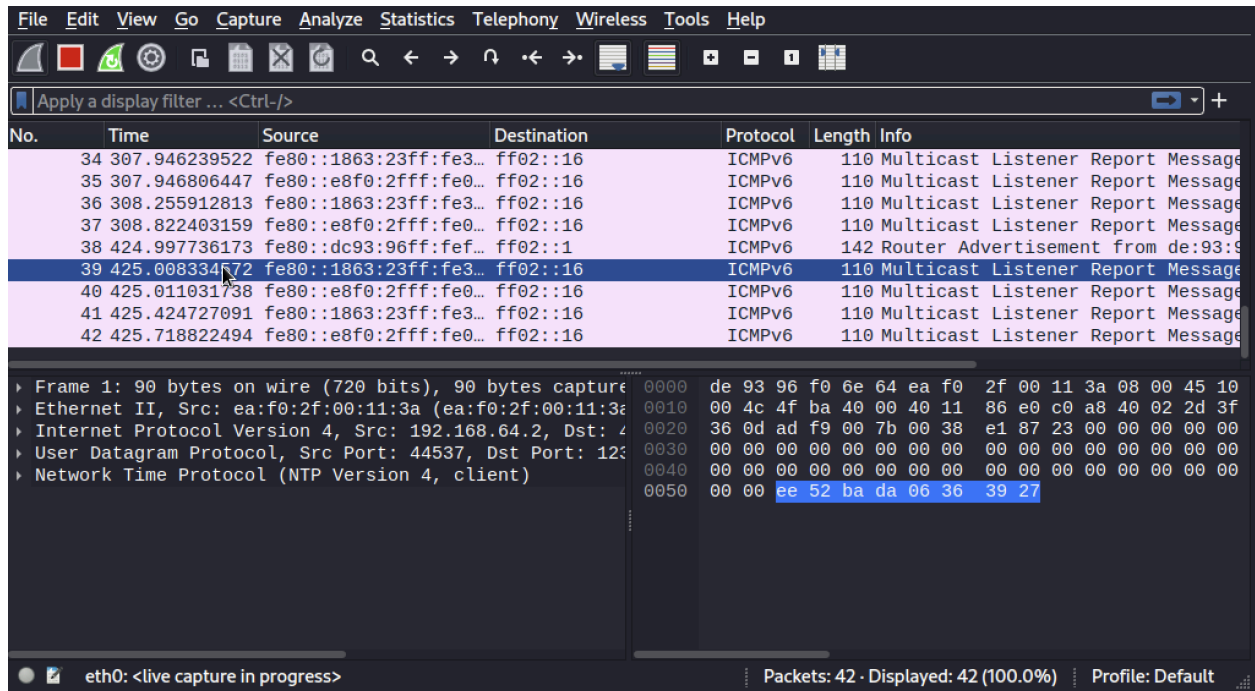
ip addr

to help identify the interface.



For now, I need to practice the easy tasks on wireshark and should not worry about advanced filters yet. That's primarily the focus of Day 4.

The capture is on the interface eth0, and it's live



To stop capturing and do analyzing, we press the red tab on the top left.

6. Prepare a capture environment

Before capturing traffic, verify:

- Kali can communicate with Ubuntu.
- Ubuntu is connected to the isolated lab network.
- Wireshark can access the appropriate interface.
- The correct interface shows network activity.
- You have a location prepared for saving **.pcapng** capture files.

For example, you could create a project capture directory:

```
mkdir -p ~/cybersecurity-lab/captures
```

What to document

For your Day 3 evidence, I would capture screenshots of:

1. **Wireshark installation** or successful launch.
2. Kali's **ip addr** showing the lab interface and **192.168.64.2**.
3. Wireshark showing the available network interfaces.
4. The selected interface ready for packet capture.
5. Your capture directory, if you create one.

Day 3 vs. Day 4

There's an important distinction:

Day	Main purpose
Day 3	Install, configure, and prepare Wireshark
Day 4	Learn interfaces, display filters, coloring rules, and practice capturing traffic

So **Day 3 should end with Wireshark fully installed and ready to capture**. And no need to do extensive packet analysis yet.